

Input

1 Corpus assembly and fittability filtering

Retrieval, deduplication by DOI, screening against the applicability window of Eq. (1)

934 screened articles

2 Curve-level extraction

Vision-assisted plot reading under a cross-model agreement gate; deterministic code thereafter

curve records: sample form, sample identifier,
fixed-axis value, source paper

3 Critical-scale resolution

Assign T_c and $H_{c2,0}$ to each fitted curve
through the provenance hierarchy of Sec. II.B

T_c and $H_{c2,0}$ per curve, each tagged with its provenance tier

4 Per-axis exponent fitting

Fit Form 3, Eq. (1), independently on each axis
within physically grouped curves

β_T , β_H , $\log_{10} J_{c, \text{partial}}$ per curve

5 Scope selection and aggregation

Variance diagnostic selects Stage 2 or Stage 3;
medians taken within the selected cell

family median parameters and the regime label

6 Dispatch through refusal gates

Anchor count, monotonicity, family population,
anchor availability, target inside calibration

refusal code
no value emitted

family-scope envelope with a 95% bootstrap interval

Predictor output